

Data Analytics Capstone Project

Global Development Indicators: An Analysis of Health, Economic, and Education Trends (1960–2024)

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1. Objective / Problem Statement

The goal of this project is to apply a complete data analytics workflow — from raw data acquisition through cleaning, analysis, visualization, and reporting — to a real-world, globally relevant dataset. Specifically, this project examines how key indicators of health, economic development, and education have evolved across countries between 1960 and 2024, using the World Bank's World Development Indicators (WDI) dataset.

Two central questions guide this analysis: first, how are economic prosperity, healthcare investment, and life expectancy related across countries, and second, what progress has been made toward global education and gender equality goals, particularly in adult literacy. The intention is not simply to visualize data, but to surface findings that could meaningfully inform decisions by policymakers, NGOs, and development organizations.

2. Dataset Description

The dataset used is the World Development Indicators (WDI), published by the World Bank, one of the most comprehensive publicly available sources of global development statistics. It covers more than 200 countries and territories across multiple themes, including economy, poverty and inequality, education, health, environment, infrastructure, trade, technology, and governance, with historical data spanning from 1960 to 2024.

From this large dataset, nine indicators were selected for focused analysis, spanning three broad themes:

- Economy: GDP growth (annual %), GDP per capita (current US\$)
- Health: Current health expenditure (% of GDP), current health expenditure per capita (current US\$), life expectancy at birth (total years), mortality rate for infants and under-5 children (per 1,000 live births)
- Technology: Individuals using the internet (% of population)
- Education: Adult literacy rate, total, and later expanded to include female and male adult literacy rates and the youth Gender Parity Index (GPI) for a deeper analysis of gender-based education gaps

3. Data Cleaning Process

The raw WDI file was cleaned and transformed using Power Query within Power BI. The following steps were applied, in order:

- Removed blank rows and duplicate rows to eliminate redundant or incomplete records.

- Filtered the dataset down to only the nine (later expanded to twelve) indicators relevant to this analysis, out of the thousand-plus indicators available in the full WDI file.
- Promoted the first row of the dataset to serve as proper column headers, since the raw file initially loaded with generic column names.
- Unpivoted the year columns: the original file stored each year (1960–2024) as a separate column, resulting in an extremely wide table. This was transformed into two columns, 'Year' and 'Indicator Value', producing a long-format table that is far better suited to trend analysis and visualization.
- Renamed columns for clarity, including 'Country Name', 'Country Code', 'Indicator Name', and 'Indicator Value'.
- Created DAX measures to calculate averages for each indicator, filtered by its exact Indicator Name, enabling clean, reusable calculations across multiple visuals.

This cleaning process converted a large, wide, and unwieldy spreadsheet into a compact, well-structured table suitable for interactive analysis and dashboard development.

4. Analysis Methodology

The analysis followed a structured approach: first identifying relevant indicators for each theme, then building calculated measures using DAX to summarize values across countries and years, and finally applying filtering and ranking techniques (Top N and Bottom N) to identify leading and lagging countries for each metric.

To explore relationships between variables, a scatter chart was built comparing GDP per capita against life expectancy across countries, allowing visual inspection of whether wealthier countries consistently report longer life expectancy, and whether that relationship is linear or shows diminishing returns. A similar comparative approach was used in the education analysis, examining literacy rates disaggregated by gender to assess the presence and size of any gender gap.

5. Key Findings

5.1 Global Health & Development Dashboard

- GDP per capita has shown an overall upward trend across the study period, although growth varies significantly between countries.
- Internet adoption has increased substantially since the early 2000s, reflecting rapid global digital transformation.
- Countries with higher GDP per capita generally report higher life expectancy, suggesting a positive relationship between economic prosperity and health outcomes; however, this relationship shows diminishing returns — life expectancy rises sharply at lower income levels and then plateaus at higher income levels.

- Health expenditure per capita differs considerably across countries, highlighting significant variation in healthcare investment.
- Some of the wealthiest countries by GDP per capita, such as Monaco and Liechtenstein, have incomplete reporting for indicators such as life expectancy, limiting full cross-country comparison for very small economies.

5.2 Education & Gender Equality Dashboard

- The global average adult literacy rate is 76.01%, meaning roughly one in four adults worldwide still cannot read or write.
- A clear gender gap persists: male adult literacy averages 81.61%, compared to 70.53% for females, a difference of just over 11 percentage points.
- The youth Gender Parity Index (GPI) stands at 0.92, confirming this imbalance; a value below 1.0 indicates young women remain slightly behind young men in literacy, though the gap among youth is narrower than among adults overall, suggesting gradual generational progress.
- At the current global average of 76.01%, literacy remains approximately 14 percentage points short of the commonly cited 90% global literacy benchmark.
- The lowest-performing countries by adult literacy rate — including Chad, South Sudan, and Mali — report literacy rates well below the global average, highlighting sharp regional inequalities that the overall average does not fully capture.

6. Insights

Taken together, the two dashboards reveal a consistent theme: while global averages for economic growth, technology adoption, and life expectancy have generally improved over the past six decades, these gains are unevenly distributed. Wealthier countries continue to see marginal health returns from additional income, suggesting that beyond a certain threshold, further gains in life expectancy require investment beyond GDP growth alone — likely in areas such as healthcare infrastructure, sanitation, and preventative care.

In education, the persistent gender gap in adult literacy, despite a narrower gap among youth, suggests that policy interventions targeting girls' education over the past two to three decades are producing measurable results, even though full parity has not yet been reached. The concentration of the lowest literacy rates in a small number of countries indicates that global averages can mask severe localized gaps that require targeted, rather than broad, intervention.

7. Recommendations

- Policymakers and development organizations should prioritize continued investment in girls' and women's education, particularly in the countries identified as having the lowest adult literacy rates, since this is where measurable gaps remain widest.

- Health policy should shift focus, in higher-income countries, from GDP-driven growth toward targeted healthcare investment, given the diminishing returns observed between GDP per capita and life expectancy at higher income levels.
- Given the narrowing gender gap among youth, sustained investment in early and secondary education for girls should continue, as current trends suggest this is a key lever for closing the adult literacy gap within the next generation.
- Further analysis could incorporate additional indicators, such as school enrollment rates and education expenditure as a percentage of GDP, to better understand the drivers behind literacy outcomes in the lowest-performing countries.

8. Conclusion

This capstone project demonstrates the application of a complete data analytics workflow, from raw data acquisition and cleaning through analysis, visualization, and reporting, using a real-world global development dataset. The resulting two-part Power BI dashboard set surfaces meaningful, evidence-based insights into the relationship between economic development and health outcomes, as well as persistent gaps in global literacy and gender equality in education. These findings reinforce that while global development indicators broadly trend in a positive direction, targeted, data-informed intervention remains necessary to close the gaps that averages alone can obscure.